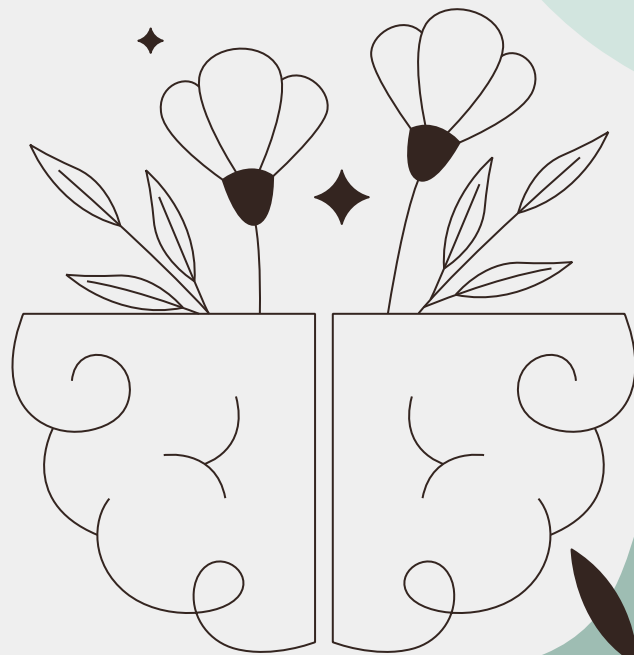




Predicting Depression Score

Based on Demographics, Income,
Weight, Working Hours, and
Sleep

By Aiden Brown and Tony Nguyen



Why This Topic?

Background:

From Aug, 2021 - Aug, 2023, the CDC reported that 13.1% of U.S. adolescents and adults over the age of 12 struggle with depression, the highest prevalence was among teenagers and adolescents.

We wanted to determine if and how **factors** such as:

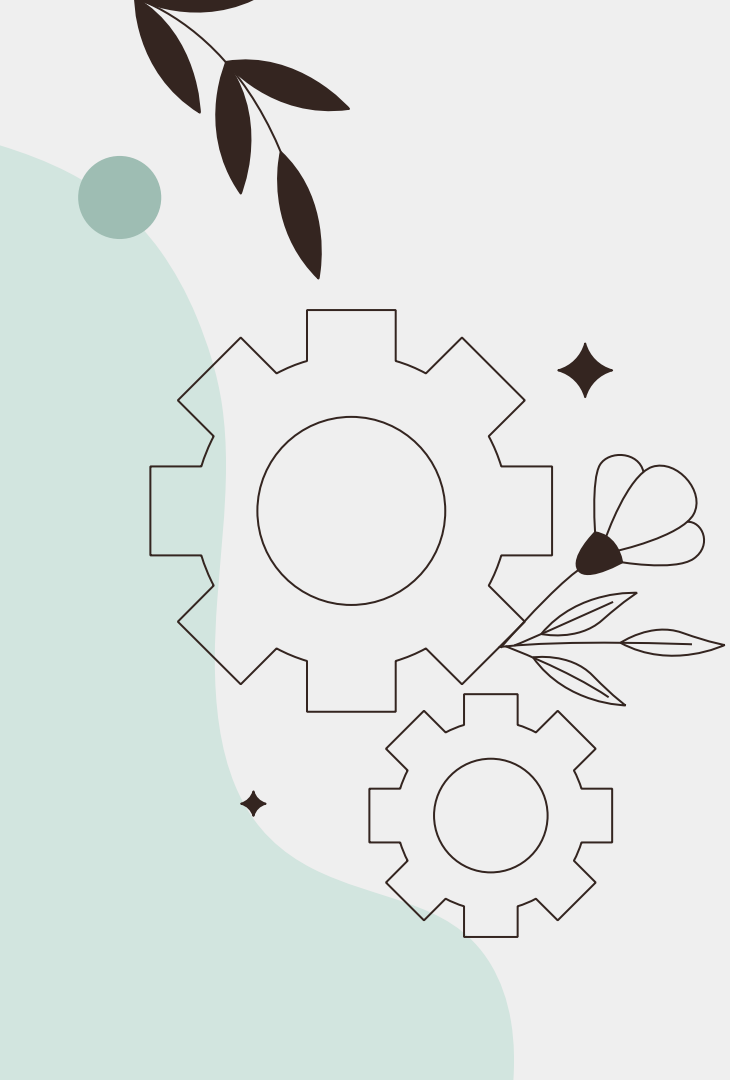
- income
 - sleep
 - demographics
 - hours worked each week
- correlate to depression diagnoses.





Research Question

What impact, if any, do different demographics, sleep behaviors, incomes, body weights, and hours worked each week have on predicting depression score?



Data Source - CDC NHANES

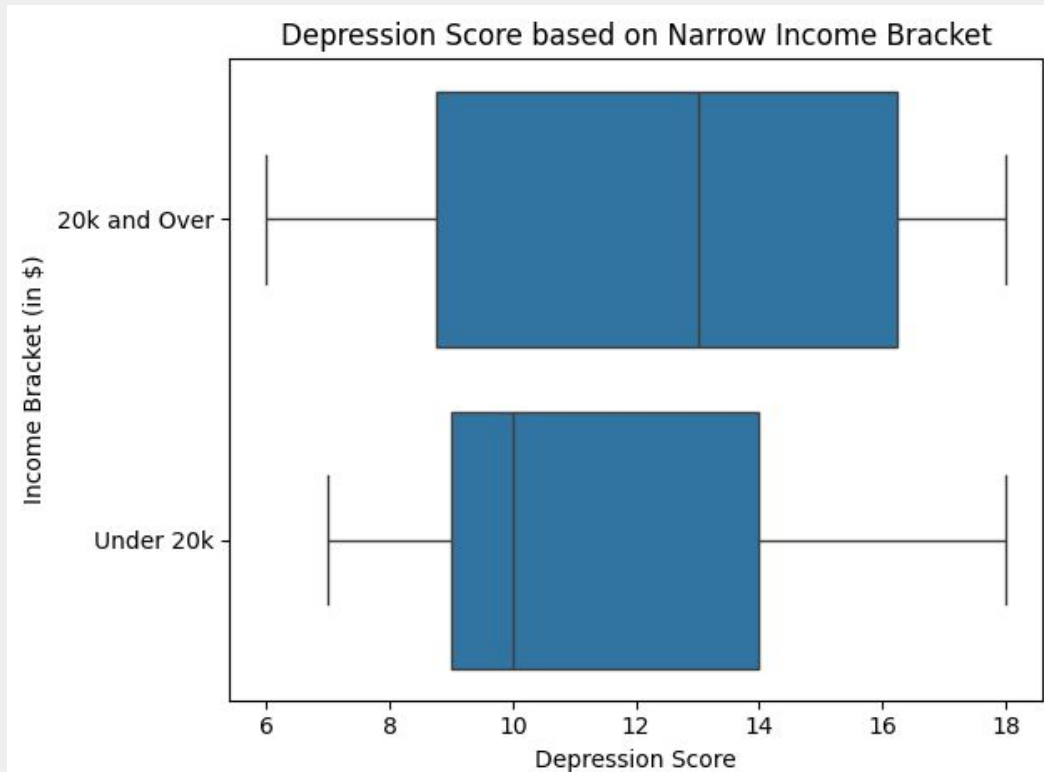
- The CDC has annual reports on several different health conditions
 - The 2017-2018 data was most accessible
 - NHANES - National Health and Nutrition Examination Survey
 - The CDC is one of the primary authorities on health in the US
 - Depression score is an aggregation of the severity of depression symptoms
- 

Data Cleaning and Wrangling Methods



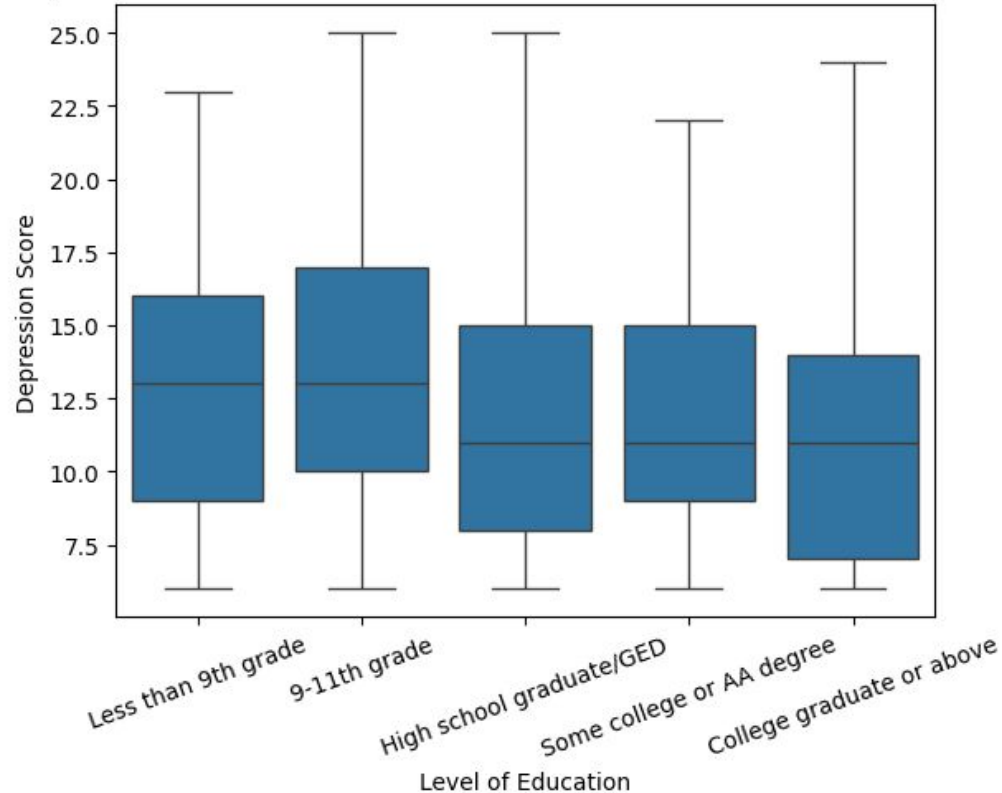
- Cleaning:
 - Replaced invalid values for depression indicators with NaN
 - Converting variables to numeric due to file conversion
 - Replaced encoded values with NaN
- Wrangling
 - Merging 4 datasets by sequence number
 - Dropped observations with less than 6 out of 9 depression indicator questions not answered
 - Dropped observations that did not have a valid depression score

Insight from EDA: Unexpected Relationship Between Income and Depression Score



Insight from EDA: Relationship Between Level of Education and Depression Score

Depression Score based on the Education Level of Adults 20 Years or Older



Predictive Models

Linear Regression & LASSO

Setup & Features:

- Trained on 5 predictors: Race, Gender, Poverty Ratio, and Weekday Sleep, and Weekend Sleep.
- LASSO utilized 10-fold cross-validation.
- Applied standardization to all variables.

Neural Network

Architecture:

- Trained for 50 epochs using the same 5-feature subset.
- Two linear layers with ReLU activation.
- Implemented a 40% dropout rate for regularization.

Gradient Boosting

Dual-Model Approach:

- Regressor for continuous depression scores.
- Classifier for binary target (Score ≥ 10).
- XGBoost models used all features with one-hot encoding.

Results of Regression Models/Comparison

1	Linear Regression	RMSE: 4.39 R ² : -0.034
2	LASSO	RMSE: 4.39 R ² : -0.034
3	Neural Network	RMSE: 4.6 R ² : -0.15
4	Gradient Boosting	RMSE: 4.48 R ² : -0.022

Overall Interpretation:

All models yielded negative R² values, indicating they performed worse than predicting the mean. The low R² suggests that the selected predictors (Race, Gender, Poverty Ratio, Sleep) may not meaningfully capture the correlation to depression scores in this dataset.

The Gradient Boosting model showed a slight improvement in R² compared to other models, suggesting a potential for other variables to be explored further.

Results of Regression Models/Comparison

1	Linear Regression	MAE: 3.60 R ² : -0.034
2	LASSO	MAE: 3.60 R ² : -0.034
3	Neural Network	MAE: 3.60 R ² : -0.034
4	Gradient Boosting	RMSE: 4.48 R ² : -0.022

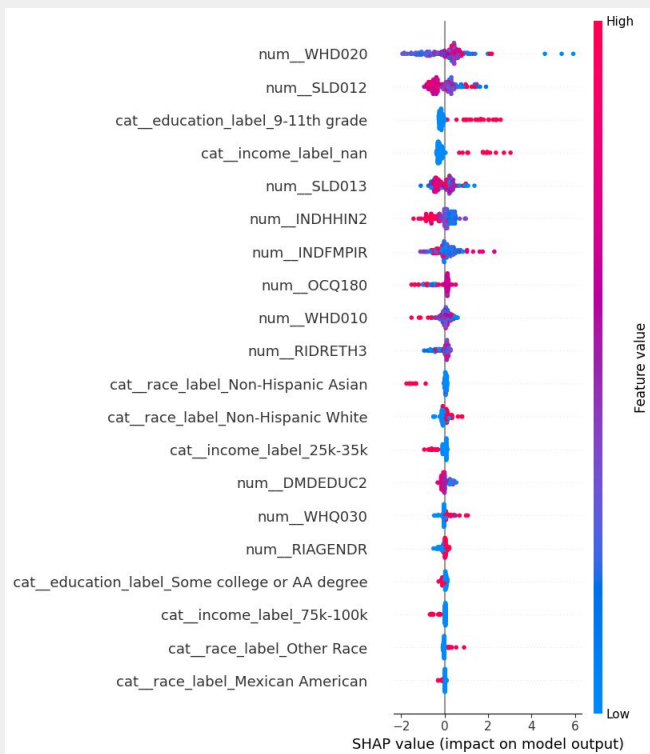
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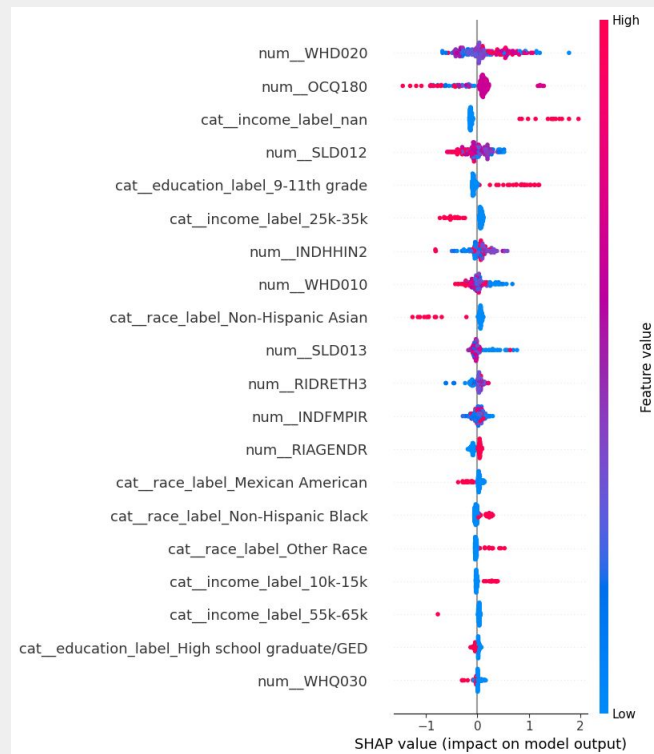
However, the Neural Network showed a slight improvement in R² compared to simpler linear models, suggesting a potential for non-linear relationships to be explored further.

SHAP Value Analysis

Feature Impact on Regression vs. Classification Models



Regression Model



Classification Model (Depression > 10)

Limitations

**1. Only used Data from US
2017-2018**

**2. We selected predictor variables
based on our hypotheses**

**3. Depression Score is not
Depression Diagnosis**

**4. Data we (necessarily) dropped
may have changed relationships**

Conclusions and Future Work

Conclusions

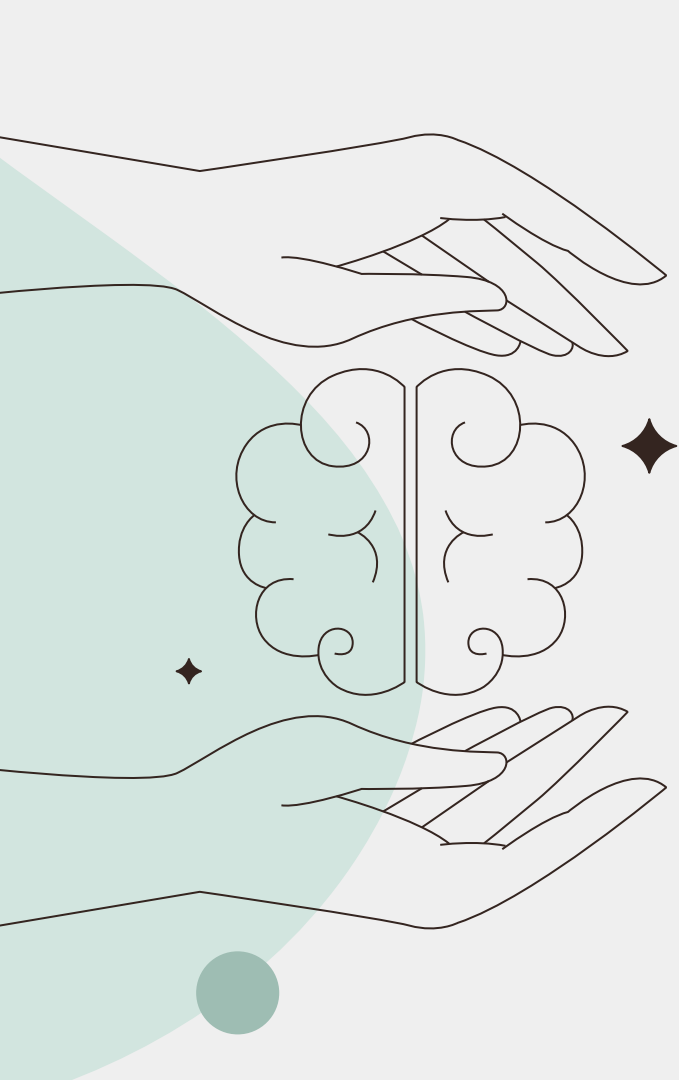


- EDA suggested possible correlations between predictors and depression score but our models had weak performance
- Our predictor variables were not strong enough on their own to predict depression score

Future Work



- How will our models perform with data from other years and other nations?
- Increase quantity and variety of predictor variables
- Compare depression score and another metric of depression? Are the same predictors relevant?



Thanks!

What questions do you have?

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